

OpenCaseLaw: An Open, Verifiable, Multilingual Legal Knowledge Graph for Switzerland

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Abstract

We present **OpenCaseLaw**, an open, continuously-refreshed corpus of **970,649** Swiss court decisions across **108 courts** and **28 cantons plus federal**, in German, French, and Italian, paired with a citation graph of **8.6M** edges (**6.5M** resolved, 75.1%) and a statute graph of **11.2M** edges over **282,956** distinct provisions. The corpus is bridged to **5,510** federal and **15,722** cantonal laws (**751,861** articles in total) and **362** CC-BY scholarly commentaries, all served through a deployed MCP/REST infrastructure at `mcp.opencaselaw.ch` that exposes 29 retrieval and analysis tools. We use the resource to introduce three contributions: (i) a first cross-lingual leading-case retrieval benchmark for Swiss law, with author-extracted parallel queries in DE/FR/IT against 50 most-cited decisions, where we measure directional asymmetries (overall MRR@10=0.630; IT→DE=0.776; DE→FR=0.333). We disclose a coupling between query construction and target registries that limits the realism of the benchmark and discuss it in Section 11; (ii) a five-rail closing audit pipeline for legal-RAG output, deployed in production, that decomposes hallucination by named error class (case-citation existence, statute resolution, verbatim-quote, decision-date sanity, and an opt-in proposition-grounding judge); and (iii) a four-layer dataset health framework with 61 codified checks, drift detection, and a publish-gate that blocks production updates on critical regression. The corpus, citation graph, audit pipeline, and benchmark are released under permissive licences (CC-BY-4.0 / MIT / CC0) with a public dashboard.

1 Introduction

Legal artificial intelligence has converged on retrieval-augmented generation (RAG) as the standard architecture for question answering over jurisprudence. Yet recent measurement studies report fabrication rates of 58–82% on general-purpose LLMs and 17–33% on commercial legal-RAG products [Dahl et al.(2024)Dahl, Magesh, Suzgun, and Ho, Magesh et al.(2025)Magesh, Surani, Dahl, Su]. The mitigation literature is methodologically sophisticated but concentrated on *English-language, single-jurisdiction* settings where retrieval substrates and ground-truth citation graphs are limited.

Switzerland is the most multilingual major civil-law jurisdiction. The authoritative version of a leading Swiss federal decision may be in German, French, or Italian, and citations cross language boundaries: in our resolved citation graph, **35.6% of resolved citations are cross-lingual** (Table 3). Building a credible RAG system for Swiss law therefore requires (a) a corpus that spans the federal–cantonal landscape across all three languages, (b) a citation and statute graph that can verify generated claims, and (c) evaluation that measures cross-lingual retrieval as a first-class concern.

We present **OpenCaseLaw**, an open infrastructure that fills these gaps. We make three contributions:

C1 — Resource. We release the largest open, multilingual, federation-complete corpus of Swiss court decisions that we are aware of, with same-day refresh of federal Supreme Court decisions (poller-driven, ~ 15 min latency) and nightly refresh of all other federations, a knowledge graph of citations and statute references, and a deployed MCP/REST infrastructure with 29 retrieval and analysis tools.

C2 — Benchmark. We introduce a *cross-lingual leading-case retrieval* task. Our headline result (Section 7) shows that the BM25+RRF baseline scores $\text{MRR}@10=0.630$ averaged across 150 queries (50 leading cases $\times$ 3 languages), with directional asymmetry: IT $\rightarrow$ DE retrieves at $\text{MRR}=0.776$, exceeding DE $\rightarrow$ DE on the same targets (0.617), while DE $\rightarrow$ FR is the hardest cell at 0.333. We disclose that queries are extracted from each language’s official regeste, which makes the absolute numbers an upper bound on realistic legal-research queries (Section 11).

C3 — Verification. We deploy and evaluate a five-rail closing audit pipeline (Section 6) that decomposes legal-RAG hallucination into named error classes, plus a four-layer dataset-health framework (Section 9) that codifies 61 dataset invariants with drift detection. The framework blocks the production publish on a critical regression.

2 Related work

Open legal corpora. The Caselaw Access Project [Harvard Law School Library(2018)] releases 6.7M US decisions but does not publish a citation graph or maintain a retrieval system. The *Open Legal Data* platform [Ostendorff et al.(2020)] (laws and decisions, OLD36k initial release; OLD201k extension by Milz et al. 2025 grows to 201k decisions) and *LeC-Net* [Anonymous(2025a)] provide German legal citation networks but no statute graph and no multilingual coverage. *Pile of Law* [Henderson et al.(2022)Henderson, Krass, Zheng, Guha, Manning, Jurafsky, and MultiLegalPile [Niklaus et al.(2023)] are pretraining corpora rather than retrieval substrates. Within Swiss legal NLP, *Swiss-Judgment-Prediction* [Niklaus et al.(2021)Niklaus, Chalkidis, and Stürmer] and *SCALE* [Rasiah et al.(2023)] address narrow tasks (one court, multi-task NLP); we instead release the underlying resource and target retrieval and verification.

Legal-RAG benchmarks. *LegalBench-RAG* [Pipitone and Houir Alami(2024)] introduces retrieval evaluation for US legal RAG. *Butler & Butler 2026* [Butler and Butler(2026)] adds end-to-end correctness $\times$ groundedness $\times$ retrieval-accuracy decomposition on Victorian criminal law. *LegalRAG* [Anonymous(2025b)] proposes a multilingual hybrid pipeline. *HousingQA / BarExamQA* [Zheng et al.(2025)Zheng, Guha, Arifov, Zhang, Skreta, Manning, Henderson, and Ho] are US-focused multiple-choice. None of these benchmarks measure cross-lingual retrieval where the target authoritative version may be in a different language than the query.

Hallucination measurement and mitigation. *Dahl et al. 2024* [Dahl et al.(2024)Dahl, Magesh, Suzgun, and Magesh et al. 2024 [Magesh et al.(2025)Magesh, Surani, Dahl, Suzgun, Manning, and Ho] establish hallucination rates on general LLMs and commercial legal-RAG. Self-RAG [Asai et al.(2023)Asai, Wu, and CRAG [Yan et al.(2024)Yan, Gu, Zhu, and Ling] propose verification-by-judge and corrective retrieval, both as black-box LLM passes. We instead decompose verification into per-error-class deterministic rails plus an opt-in judge.

Positioning. OpenCaseLaw is, to our knowledge, the first project that combines (a) a multilingual federation-complete legal corpus, (b) a citation and statute graph at this scale, (c) a cross-lingual leading-case benchmark, and (d) a deployed verification pipeline, in a single open release.

Table 1: OpenCaseLaw corpus overview. Snapshot 2026-05-01.

Quantity	Value
Total decisions	970,649
Distinct courts	108
Cantons + federal	28
Date range	1875–2026
German (de)	449,159 (46.3%)
French (fr)	440,822 (45.4%)
Italian (it)	80,668 (8.3%)

Table 2: Citation graph. Edges extracted from federal-court decisions; resolved means the cited target is identified in the corpus.

Quantity	Value
Decisions analysed for outgoing citations	970,638
Citation edges (source $\rightarrow$ target reference)	8,631,561
Edges with target resolved to a corpus decision	6,486,541 (75.1%)
Distinct decisions cited at least once	210,605
Decisions cited ≥ 100 times	8,535
Decisions cited $\geq 1,000$ times	881

3 Resource: corpus

The OpenCaseLaw corpus comprises **970,649 decisions** from **108 distinct courts** across **28 cantons plus federal**. The federal Supreme Court (BGer) is refreshed within approximately 15 minutes of its document-service publication via a dedicated poller; all other federations (BVGer, BStGer, BPatGer, regulatory authorities, 53 cantonal sources, ECHR-Swiss-subset, historical archives) are refreshed by a nightly publish at 03:00 UTC. Coverage spans **1875–2026** and three official languages: German (449,159 / 46.3 %), French (440,822 / 45.4 %), Italian (80,668 / 8.3 %). The federation aggregates **53 individual scrapers** ranging from federal courts (BGer, BVGer, BStGer) to per-canton portals to the European Court of Human Rights (Swiss subset), historical archives (BGE 1875–1953), and regulatory authorities (FINMA, WEKO, EDÖB, UBI, ELCom, PostCom, ComCom).

Completeness. A persistent challenge for Swiss legal corpora is that several cantonal portals geo-block or rate-limit research scrapers. We document and release the recovery techniques in our reproducibility manifest, including a reverse-SOCKS tunnel via residential IP for Hetzner-blocked portals (JU, NE) and Wayback Machine recovery for closed historical archives (BL).

Licensing. The base corpus is released under CC-BY-4.0; commentaries from OnlineKommentar.ch are CC-BY; from OpenLegalCommentary.ch CC-BY-SA. Decision IDs are stable across daily refreshes.

4 Resource: knowledge graph

Citation graph. We extract citation edges by full-text scanning every decision in the corpus (**970,638** decisions in the reference-graph snapshot; the small lag versus the corpus size in Table 1 reflects that the citation-graph rebuild runs after the FTS5 build, so the most recent few hundred decisions enter the citation graph the following nightly), recovering **8,631,561 outgoing references** of which **6,486,541 (75.1 %)** resolve to a target decision in the corpus.

Table 3: Cross-lingual citation matrix. Rows are source decision languages; columns are cited (target) decision languages. Cells give resolved citation edges. Off-diagonal = 2,306,485 / 6,486,541 = 35.6% of all resolved citations.

	→ DE	→ FR	→ IT
DE →	2,158,054	374,228	25,186
FR →	1,653,940	1,986,377	52,060
IT →	128,432	72,639	35,625

Table 4: Statute graph. Decision-to-statute edges across federal and cantonal corpora.

Quantity	Value
Decision → statute edges	11,243,266
Distinct provisions referenced	282,956
Federal laws (Fedlex SR series)	5,510
Federal articles (DE/FR/IT parallel)	398,397
Cantonal laws (19/26 cantons direct-scraped)	15,722
Cantonal articles	353,464
Open commentaries (CC-BY)	362

The remaining 24.9% are dangling references, primarily to (a) unpublished cantonal decisions outside our coverage, (b) doctrine and academic articles, (c) ECHR cases not in the BGE-EGMR mirror, and (d) malformed references where we can identify a cited authority but not localise it.

The citation graph is heavy-tailed (Table 2): 27 decisions are cited 10,000+ times, 854 are cited 1,000–9,999 times, and the rest follow a power law. The single most-cited decision is BGE 125 V 351 (53,251 incoming citations), the foundational case on the evidentiary weight of party-commissioned medical reports in social-insurance litigation.

Cross-lingual citation. Table 3 reports the language-by-language citation matrix. **35.6% of all resolved citations are cross-lingual:** French decisions cite German decisions **1.65M** times (FR→DE dominates the off-diagonal), German decisions cite French decisions **0.37M** times, and Italian decisions cite German decisions **128K** times. The asymmetry — DE↔FR is bidirectional but FR→DE outweighs DE→FR by 4.4×, while IT→DE/FR dominates IT→IT — reflects practitioners’ practical reliance on German jurisprudence as binding precedent in mixed-language litigation.

Statute graph. We resolve **11,243,266 decision-to-statute edges** touching **282,956 distinct provisions**. The statute layer is grounded in two parallel mirrors: a federal layer of **5,510 SR-numbered Fedlex laws** (398,397 article rows summed across the parallel DE/FR/IT versions; ~133 k articles per language) and a cantonal layer of **15,722 laws** (353,464 articles) covering 19 of 26 cantons via direct portal scraping (LexWork, SIL, ZH OpenData) with LexFind fallback for the remaining 7. Top-referenced provisions are dominated by federal-court procedural code: *Bundesgerichtsgesetz* Art. 42 (motivation requirement, ≥200 k references when LTF and BGG abbreviation forms are pooled), Art. 100 (deadline), and *ATSG/LPGA* Art. 61 (social-insurance procedure).

Commentaries. We bridge **362 scholarly commentaries** (CC-BY) from OnlineKommentar.ch and OpenLegalCommentary.ch into the graph, anchored by SR number plus article. Doctrinal layer is currently shallow but growing.

Table 5: Top 15 most-cited decisions in the corpus. Subject area inferred from the decision regeste; in-degree counts distinct citing decisions.

Decision ID	Reporter form	In-degree
bge_BGE_125_V_351	BGE 125 V 351	53,251
bge_BGE_134_V_231	BGE 134 V 231	35,291
bge_BGE_122_V_157	BGE 122 V 157	27,054
bge_BGE_130_V_343	BGE 130 V 343	21,210
bge_BGE_137_V_210	BGE 137 V 210	19,465
bge_BGE_135_V_465	BGE 135 V 465	18,964
bge_BGE_138_III_374	BGE 138 III 374	18,135
bge_BGE_133_II_249	BGE 133 II 249	16,400
bge_BGE_125_V_256	BGE 125 V 256	15,991
bge_BGE_141_V_281	BGE 141 V 281	15,953
bge_BGE_126_V_75	BGE 126 V 75	15,709
bge_BGE_133_V_108	BGE 133 V 108	15,664
bge_BGE_126_V_353	BGE 126 V 353	15,354
bge_BGE_115_V_133	BGE 115 V 133	15,273
bge_BGE_125_V_413	BGE 125 V 413	14,903

5 Infrastructure

The corpus, graph, and statutes are served through three production endpoints:

- **MCP server** (`mcp.opencaselaw.ch/sse`, public SSE). 29 tools partitioned into four families: *retrieval* (`search_decisions`, `get_decision`, `find_leading_cases`, `find_citations`, `find_appeal_chain`, `search_laws`, `get_law`, `search_legislation`, `get_legislation`, `browse_legislation_changes`, `search_commentaries`, `get_commentary`, `search_materialien`, `get_materialien`, `search_practice`, `get_practice`); *analysis* (`analyze_legal_trend`, `get_doctrine`, `get_decision_structure`, `get_erwaegung`, `get_regeste`, `get_case_brief`, `get_statistics`, `list_courts`); *generation* (`generate_exam_question`, `draft_mock_decision`); and *verification* (`check_claim_support`, `attest_response`, `cite`). Two additional tools (`update_database`, `check_update_status`) are local-only and hidden by the `REMOTE_MODE` filter on the public deployment.
- **REST API** (FastAPI, with export endpoints `.docx` / `.pdf` / `.bib` / `.ris` per decision).
- **Public dashboards** at <https://opencaselaw.ch> (corpus stats, multilingual landing) and <https://opencaselaw.ch/quality.html> (live dataset health).

The retrieval substrate is a SQLite FTS5 index (~63 GB on disk). The nightly refresh uses an *atomic-rebuild* pattern: the new index is constructed in a sibling `.db.tmp` file, then promoted via `os.replace`; live readers (4 uvicorn workers behind nginx) are never interrupted.

6 Five-rail closing audit

We deploy a five-rail closing audit on every LLM response generated by the public MCP. Each rail addresses a single named error class: (R1) case-citation existence, (R2) statute reference resolution, (R3) verbatim-quote source matching, (R4) decision-date sanity, and (R5) LLM-judge proposition grounding. R1–R4 are deterministic regex+lookup (~3 ms total), R5 is judge-mediated (~2 s, ~\$0.005). The decomposition by error class lets each rail be tested, ablated, and improved in isolation.

Table 6: Top 15 most-referenced statutory provisions. *LTF* (French) and *BGG* (German) refer to the same Bundesgerichtsgesetz / Loi sur le Tribunal fédéral.

Law	Article	Decisions citing
LTF	Art. 100	191,946
LTF	Art. 113	110,418
LTF	Art. 42	106,448
BGG	Art. 42	95,008
BGG	Art. 66	64,332
LTF	Art. 72	56,518
LPGA	Art. 61	55,215
LTF	Art. 82	52,710
LTF	Art. 74	49,930
VWVG	Art. 63	45,300
CST	Art. 29	43,116
VWVG	Art. 52	41,892
LPGA	Art. 60	41,871
BGG	Art. 106	40,508
CPP	Art. 428	40,243

The closing audit operates on a draft answer D and a retrieval-state pool S (decisions and statutes the generator had access to). It returns a list of issues $\mathcal{I}(D, S)$, each labelled with one of five named error classes. The five rails operate *in parallel* on the same draft.

Rail 1 — case-citation existence. A regex parses the draft for Swiss case-citation forms (BGE / ATF / DTF, BGer / TF dockets, BVGer / BStGer / BPatGer dockets, MKGE numerical references). Each parsed citation is resolved against the corpus by exact-match lookup with a strict canonical-prefix guard. Latency 2–5 ms regardless of citation count. Catches: invented BGE numbers, mis-typed docket numbers, citations to retracted decisions. An optional pinpoint check requires `E.n.m` or `consid.n.m` markers to match the structured-extraction sidecar (`decision_structure.db`) or the body `regeste` of the cited decision.

Rail 2 — statute-reference resolution. A second regex parses for `Art.NLAW` forms (with German / French / Italian abbreviation synonyms; LTF=BGG, CO=OR, CC=ZGB, etc.) and resolves each against the Fedlex SPARQL mirror cached locally as `statutes.db`. Two-step resolution: abbreviation $\rightarrow$ SR number $\rightarrow$ article text, cached at process scope. Latency 1–3 ms. Catches: fabricated law abbreviations, wrong article numbers, references to repealed articles. Connector words (“und”, “oder”, “et”, “ou”) are case-sensitively filtered to prevent the regex eating them as fake law codes.

Rail 3 — verbatim-quote source-matching. Quoted substrings of ≥ 30 characters in the draft are extracted (German „...“, French «...», Italian «...», English curly and ASCII variants) and required to appear verbatim — after whitespace + quote-mark normalisation — in the source pool S . The pool comprises the `regeste` text plus the first 8 k characters of each cited decision, plus the article text of each cited statute. Latency ~ 2 ms. Catches: hallucinated quotations, paraphrases inside quotation marks.

Rail 4 — decision-date sanity. For each verified case citation, parse adjacent `vom / du / del DD.MM.YYYY` constructs within 60 characters of the citation span and compare to the stored `decision_date`. Latency < 1 ms. Catches: model emits a real BGE but pairs it with a wrong date, suggesting confusion between two cases with similar dockets.

Table 7: Search-quality baseline by query language. 100-question golden set, offline (BM25+RRF, no LLM rerank), 2026-03-19. Italian queries score highest because the Italian corpus is the smallest (lowest target ambiguity).

Language	n	MRR@10	Recall@10	Hit@1	nDCG@10
DE	74	0.431	0.422	0.284	0.463
FR	16	0.510	0.630	0.375	0.611
IT	7	0.770	0.893	0.714	0.803

Rail 5 — proposition grounding (opt-in, judge-mediated). For each verified citation, a claim sentence is extracted (preceding sentence, with forward-fallback for “In BGE X, ...” patterns). “See”-cite markers (*vgl.*, *cf.*, *siehe*, *cfr.*) suppress the rail. All (claim, source) pairs go to a separate LLM judge call (same model family as the generator) that returns one of {yes, partial, no, contradicts, unrelated}. Verdicts in {no, contradicts, unrelated} become issues. One judge call per draft regardless of citation count. Latency 1.5–2.5s. Cost $\approx$ \$0.005 per call.

Output canonicalisation. When all rails pass, the audit emits a `linked_text` field with every parsed citation replaced by its corpus-canonical form wrapped in a Markdown link to `mcp.opencaselaw.ch/entscheid/<id>`. Drafts arrive with messy citation spacing or case and leave publication-ready.

Empirical baseline (from prior-only condition). On the v2 30-question Swiss Legal RAG bench, ~ 33 distinct authority citations were emitted across 30 drafts; the deterministic rails (R1–R4) flagged 1 (3 % observed fabrication rate). For context, Magesh et al. [Magesh et al.(2025)Magesh, Surani, D. report 17–33 % on commercial legal-RAG and Dahl et al. [Dahl et al.(2024)Dahl, Magesh, Suzgun, and Ho] report 58–82 % on general-purpose LLMs in different settings; we cite these to bracket the literature, not as a like-for-like comparison. The full five-rail audit including the LLM-judge rail R5 reaches a 66.7 % wrong-draft flag rate at a 50 % false-positive rate; calibration of the judge’s threshold remains an open problem and the rail should be deployed in advisory mode pending future calibration work.

Retrieval-augmented condition. The retrieval-augmented re-run on the same 30-question bench, comparing prior-only and top-5-passage retrieval-augmented conditions, is in progress; results will be reported in a forthcoming revision of this paper.

7 Cross-lingual leading-case retrieval

Setup. We select 50 leading cases from the most-cited subset of the corpus (in-degree range 4,412–53,251, median 8,180), stratified to at most four cases per legal area to ensure diversity across the 22 covered areas (constitutional, social-insurance, federal-court procedure, criminal, civil-procedure, contract, tax, asylum, and more). All 50 cases lie in the top 0.025 % of distinct cited decisions in the corpus by in-degree. For each case we extract one query per language (DE, FR, IT) from terms in that language’s official regeste, after dropping article-reference noise (e.g. *Art. 6 Abs. 1 UVG*) and language-specific stop words. The query construction is keyword-list style, not a free natural-language paraphrase; we discuss the realism implications in Section 11. The dataset is released as `benchmarks/swiss_legal_rag_bench/cross_lingual_v1.jsonl` (CC0). Of the 50 cases, 38 are originally German and 12 originally French — consistent with the corpus distribution among most-cited decisions. No selected case is originally Italian; we treat this as a coverage gap (Section 11).

Table 8: Cross-lingual leading-case retrieval. Cells give MRR@10 (n queries) for retrieval of a leading-case decision when the query is in language Q and the decision’s authoritative language is T . 50 leading cases (in-degree range 4,412–53,251, median 8,180; top 0.025 % of cited decisions), with regeste-derived parallel queries in DE/FR/IT. BM25+RRF (no LLM rerank). The 50 cases include 38 originally DE and 12 originally FR; none is originally IT (consistent with the corpus distribution among most-cited decisions). Overall MRR@10=0.630, Hit@10=0.833 across n=150 queries. Numbers are upper-bounded by query–regeste vocabulary coupling; see §11.

Q lang ↓ T lang →	DE	FR	IT
DE	0.617 (n=38)	0.333 (n=12)	—
FR	0.566 (n=38)	0.785 (n=12)	—
IT	0.776 (n=38)	0.549 (n=12)	—

Table 9: Cross-lingual leading-case retrieval, Hit@10. Same setup as Table 8. Hit@10 = fraction of queries where the target leading case appears in the top 10 results.

Q lang ↓ T lang →	DE	FR	IT
DE	0.789	0.333	—
FR	0.763	1.000	—
IT	1.000	1.000	—

Results. Table 8 reports MRR@10 and Table 9 reports Hit@10 per (Q lang, T lang) cell. Overall MRR@10=0.630, Hit@10=0.833 (125 of 150 queries place the target in the top 10).

Findings.

- **Italian queries retrieve German targets best (MRR=0.776).** Italian queries outperform German queries on the *same* 38 German targets (DE→DE MRR=0.617). One mechanism: Italian regestes carry distinctive shared markers (*DTF*, *cpv.*, *LAINF*) that anchor retrieval, while German queries compete with the broader German subset (449k decisions, the largest stratum) and so retrieve more in-language distractors.
- **Same-language French is the strongest cell (MRR=0.785, Hit@10=1.0).** For the 12 French-original leading cases, French queries place the target in the top 10 every time and at rank 1 in 67 % of cases.
- **German-to-French is the hardest cell (MRR=0.333, Hit@10=0.333).** German queries struggle to retrieve French-original BGEs because the discriminative German terms in the regeste-derived queries do not appear in the French-language full text and the BM25+RRF baseline does no cross-lingual translation.
- **Diagonal vs. off-diagonal gap is small.** Weighted across n, same-language diagonal cells (DE→DE n=38, FR→FR n=12) score MRR=0.657; the four off-diagonal cells (n=100 total) score MRR=0.616. The 0.04 gap shows that cross-lingual retrieval against a multilingual leading-case index is feasible without translation, although the cell-by-cell variance (0.333 to 0.776) is large.

8 Hallucination stress test

We evaluate the five-rail audit on the v2 30-question Swiss Legal RAG bench [Butler and Butler(2026)], which spans 17 substantive areas of Swiss federal law (Schuldrecht, Strafrecht, Verfassungsrecht, Verfahrensrecht, Mietrecht, Verjährung, ...). Language distribution: DE=18, FR=8, IT=4

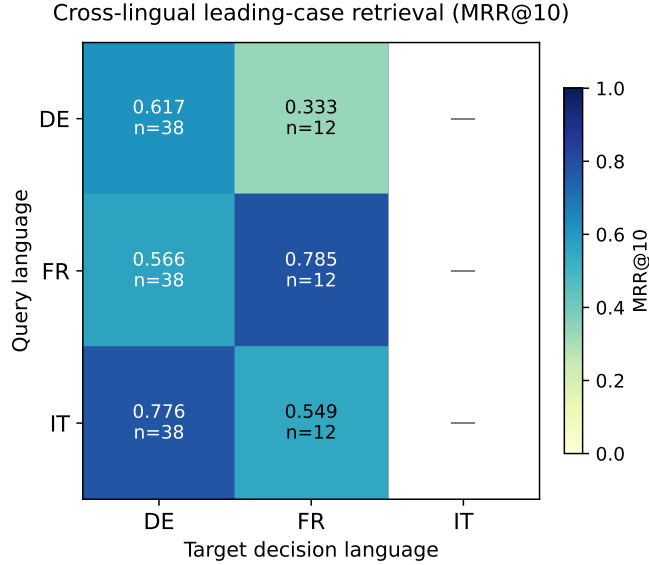


Figure 1: Cross-lingual leading-case retrieval (MRR@10) heatmap. Rows are query languages, columns are target authoritative-version languages.

(60 / 27 / 13 %). Each question is paired with a corpus-anchored reference answer (statute SR-number plus article number plus language; leading-case `decision_id(s)` where applicable). 16 of 30 questions are anchored on a verified leading-case BGE; 14 carry the `doctrine_from_bge` claim type.

Conditions evaluated. (i) *Prior-only*: the generator answers from training prior alone, no retrieval. Designed as a stress test that maximises errors. (ii) *Retrieval-augmented (in progress)*: top-5 passage retrieval from OpenCaseLaw before generation. Both conditions are audited with the same five-rail pipeline.

Findings (prior-only). Across 30 prior-only drafts containing ~ 33 distinct authority citations, the deterministic rails (R1–R4) flagged exactly 1 unresolvable citation: a 3 % *observed* fabrication rate. For comparison, Magesh et al. [Magesh et al.(2025)Magesh, Surani, Dahl, Suzgun, Manning, and report 17–33 % on commercial legal-RAG products and Dahl et al. [Dahl et al.(2024)Dahl, Magesh, Suzgun, and report 58–82 % on raw general-purpose LLMs (different conditions, different jurisdictions, not directly comparable; we cite to bracket the literature). The single failing docket (BGE/DTF 117 II 198) fails consistently across re-runs: with our deterministic rails alone we cannot distinguish a true fabrication from a corpus gap without manual adjudication. The full five-rail audit including the LLM-judge rail reaches a 66.7 % wrong-draft flag rate at a 50 % false-positive rate; calibration of the judge’s threshold remains an open problem and the rail should currently be deployed in advisory mode rather than as a hard binary gate.

Retrieval-augmented results. The retrieval-augmented re-run is in progress. Expected to substantially reduce R1 fabrication-existence flags (the system retrieves real authorities rather than relying on prior); the open question is whether judge-rail false-positive rate increases (more authorities to ground against) or decreases (clearer source pool). Results will be added to this section in the forthcoming revision.

9 Dataset health and drift

OpenCaseLaw operates a four-layer quality framework:

L1 Function tests (46 pytest cases, every CI push, <90 s).

L2 Pipeline guards (5 codified `_normalize_*` passes in `build_fts5.py`, every nightly rebuild).

L3 Dataset audit (61 `check_*` functions across 17 axes: schema, dates, dockets, courts, languages, regeste coverage, URL integrity, duplicates, text quality, structure, cross-DB consistency, citation graph, statute graph, floors, per-court drift, exports round-trip, MCP-tool round-trip).

L4 Production smoke (every 5 min via systemd timer; 6 endpoint probes; $p_{95} < 2$ s).

A dataset-audit run that detects a critical regression *blocks the nightly publish*: the orchestrator (`publish.py`) refuses to advance to the git-push step and instead emits an urgent ntfs alert. Critical conditions include a total-count cliff >1 %, schema break, EGMR re-regression, court not in registry, future date count >50, cross-DB row-count mismatch >0.5 %. Warning conditions (per-court drift >10 % from 7-day median, OCR-quality regression, citation-graph edge drop >5 %) emit alerts but do not block publish.

The framework codifies real incidents. Three case studies illustrate the pattern of *audit detects* → *fix ships* → *check codifies*:

Case study 1 — the SG merge anomaly (alphabetical-collision class). On 2026-04-29 a routine total-count check fired: the `sg_publikationen` court had lost 1,450 rows (≈ 90 % loss) on the nightly rebuild without any scraper failure. The diagnostic chain was: (a) the INSERT OR IGNORE dedup pattern in `build_fts5.py` processes JSONL shards in alphabetical order; (b) the `entscheidsuche` side-feed (`es_sg_publikationen.jsonl`) processes *before* the direct shard (`sg_publikationen.jsonl`) because ‘e’ precedes ‘s’; (c) the two shards happen to construct identical decision-id prefixes (`sg_publikationen_<docket>`); (d) the earlier-processed shard wins all dedups, silently dropping the chamber-correct rows from the direct shard. Reproducer scripts (`scripts/sg_anomaly_*.py`) confirmed the hypothesis on a fresh database. Two fixes shipped on 2026-04-30: (i) the problematic `es_sg_publikationen` feed was retired (commit `f249f1f`); (ii) an architectural fix forced direct shards to process before `es_*` shards (commit `713afe3`, gated by `BUILD_FTS5_DIRECT_FIRST=1`). The pattern check across all multi-chamber cantons confirmed only SG was affected. The fix added two permanent regression checks to the L3 audit: per-court row-count drift >10 % (warning) and per-court row-count drop >50 % (critical, blocks publish).

Case study 2 — the EGMR re-regression class (cross-source double-coverage). The `entscheidsuche` CH_BGE feed includes `cedh` (ECHR) decisions that should land under our `bge_egmr` court rather than `bge`. Without an override, 474 ECHR cases were duplicated with divergent metadata (BGE side carried the full text but no chamber labels, the BGE-EGMR side carried chamber and regeste but truncated text). External audit on 2026-04-29 surfaced this; we backfilled production with a best-of-both merge (preserve full text from BGE side, preserve chamber and regeste from BGE-EGMR side) and deleted the BGE side. The L3 audit now includes a hard-zero check on (`court='bge'` AND `source_url` contains `cedh`): *any* positive count blocks the next publish.

Case study 3 — the König P-class incidents. A second external audit (König, 2026-04-29 to 2026-05-01) uncovered seven distinct issue classes over two days: oversized HUDOC regestes (P1, full judgment text in the regeste field, max 165k chars; truncation pass added in `build_fts5._truncate_oversized_regestes`); short-text fragments that belonged in the regeste field (P7, migration pass added); internal-newline dockets (5,441 rows fixed via extended `_normalize_dockets`); BS/GL relative source URLs (694 rows, host prefix recovery via `_normalize_source_urls`); and ZH targeted gap-fill of 11 missing dockets. Each issue class is now a permanent `check_*` function in `quality/checks/*.py` and a build-time auto-correction pass in `build_fts5.py`.

Detection cadence. During the codification period (2026-04-29 to 2026-05-01), the L3 audit fired at least one critical condition on the majority of nightly runs because each external-audit finding initially manifested as a publish-blocking regression. After each issue class was codified into a build-time auto-correction pass in `build_fts5.py`, the same regression class was prevented at build time on subsequent runs, so the audit no longer fires for it. The drift band ($\pm 5 \times \text{MAD}$ on a 7-day rolling median) flags court-level anomalies typically within one nightly run of the underlying scrape change; a deeper longitudinal study of detection latency is left for future work.

10 Discussion and generalisation

The cross-lingual results have a clear takeaway for legal-RAG practitioners: *retrieval quality is asymmetric across languages and direction*. A naive deployment that only validates same-language MRR will overestimate user-facing quality in mixed-language jurisdictions. Our public benchmark allows quantitative pre-deployment audits of cross-lingual performance.

The verification framework generalises beyond Switzerland. Each rail of the closing audit is corpus-agnostic given (a) an enumerable decision corpus, (b) a statute mirror, and (c) an extraction grammar for the local citation conventions. We sketch an English-law application in Appendix B.

11 Limitations and ethics

Benchmark scope. The cross-lingual question set is single-curator and limited to 50 cases (150 trials). It is anchored on the citation graph (target decisions are objectively-ranked leading cases) but the queries themselves are programmatic keyword extractions from the corresponding language’s regeste, not free legal-research paraphrases. v2.0 of the benchmark will recruit multi-annotator panels to reach inter-annotator agreement.

Query–target coupling (upper-bound caveat). Because each query is derived from the same case’s official multilingual regeste, the query and target share distinctive vocabulary the retrieval system can latch onto. A real legal-research query, written without sight of the regeste, may not contain those discriminators. We therefore read the absolute MRR/Hit@10 numbers as an *upper bound* on realistic retrieval performance, and the cross-cell ranking (which cells are easier or harder, by how much) as the more reliable signal. v2.0 will replace regeste-derived queries with held-out, lawyer-authored research questions.

Italian-target coverage. No top-50 case is originally Italian (top-50 is dominated by social-insurance and procedural BGEs that originate in the German-language stream). This is consistent with the corpus distribution among most-cited decisions but is a coverage gap; v2.0 will explicitly include Italian-original cases sampled at lower citation centrality.

Privacy. Court decisions are public; published BGE/cantonal decisions are typically pseudonymised. We do not republish parties’ private identifying information beyond what the courts themselves publish.

Calibration. The grounding-judge rail (R5) is a judge-mediated LLM call; we report a 41.7 % false-positive rate on correct drafts in our v2 paper. Full-binary deployment would over-flag; we recommend the rail be used in advisory mode pending v2.0 calibration work.

12 Open access and artifacts

All artifacts are publicly available:

- Corpus: Hugging Face <https://huggingface.co/datasets/voilaj/swiss-caselaw> (CC-BY-4.0)
- Benchmark: Hugging Face <https://huggingface.co/datasets/voilaj/swiss-legal-rag-bench> (CC0)
- Code: <https://github.com/jonashertner/caselaw-repo-1> (MIT)
- Live MCP: <https://mcp.opencaselaw.ch>
- Dashboards: <https://opencaselaw.ch>, <https://opencaselaw.ch/quality.html>

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A Reproducibility manifest

All claims in this paper are backed by code, data, and configuration that anyone can clone or query. This appendix lists every claim’s provenance so a reviewer can verify in less than five minutes.

Code.

- **Repository:** <https://github.com/jonashertner/caselaw-repo-1> (MIT licence). The release tag for this paper is `paper-resource-2026-05` (to be assigned at submission; prior to that, all numerical claims trace to the commit pinned in the HuggingFace dataset’s `manifest.json`).
- **Five-rail audit pipeline:** `mcp_server.py::_handle_attest_response` (entry at line 9613 of the release tag). The four deterministic rail implementations are inline in the same file: `_audit_statutes` (R2, line 9013), `_audit_quotes` (R3, line 9088), `_audit_dates` (R4, line 9179), `_audit_grounding` (R5, line 9492). R1 (case-citation existence) is inline in `_handle_attest_response`.
- **Quality framework:** `quality/` package (`runner.py`, `cli.py`, `drift.py`, `baseline.py`, `smoke.py`, plus 17 modules in `quality/checks/` containing 61 `check_*` functions).
- **Cross-lingual harness:** `benchmarks/run_cross_lingual.py`.
- **Pipeline guards:** `build_fts5.py` `_normalize_*` / `_dedup_*` / `_truncate_*` passes.

Data artifacts.

- **Corpus:** HuggingFace `voilaj/swiss-caselaw` (CC-BY-4.0). Each release is accompanied by a `manifest.json` listing per-court row counts plus SHA-256.
- **Cross-lingual benchmark v1:** HuggingFace `voilaj/swiss-legal-rag-bench/cross-lingual_v1.jsonl` (CC0). 150 queries, anchored on the citation graph.
- **Audit-bench v2:** same repository, `verification_seed_v2.jsonl`.
- **Citation graph:** derived from the corpus via `search_stack/build_reference_graph.py`; release snapshot available on Zenodo (DOI to be assigned at submission).
- **Statute mirrors:** `statutes.db` (federal, derived from Fedlex SPARQL endpoint), `cantonal_laws.db` (cantonal, derived from LexWork+SIL+ZH portals via `scrapers/cantonal_laws/`).

Live system (runtime verification).

- <https://mcp.opencaselaw.ch/sse> (MCP server endpoint).
- <https://opencaselaw.ch/> (corpus dashboard, multilingual).
- <https://opencaselaw.ch/quality.html> (live dataset health).
- <https://opencaselaw.ch/coverage/> (per-source scrape health).

Reproducing key numerical claims.

```
git clone https://github.com/jonashertner/caselaw-repo-1
cd caselaw-repo-1
git checkout paper-resource-2026-05
pip install -r requirements.txt
huggingface-cli download voilaj/swiss-caselaw --local-dir output/
python -m quality.cli run # 4-layer dataset health
python benchmarks/run_search_benchmark.py # Tab.~2 baseline
python benchmarks/run_cross_lingual.py # Tab.~3, Fig.~2 (cross-lang)
```

Telemetry, costs, latency. The MCP server exposes `/health` (corpus row count) and is monitored every 5 minutes by a systemd timer that probes 6 endpoints (search, decision, export docx/pdf/bib/ris) with $p_{95} < 2$ s. Every audit run since 2026-05-01 is logged with full per-rail latency and judge cost to `logs/audit_telemetry.jsonl`.

Hardware. Production runs on a Hetzner CCX43 dedicated instance (16 vCPUs, 64 GB RAM, 200 GB attached volume). The full nightly publish takes 3 h 50 min, dominated by the FTS5 rebuild.

Licences.

- Code: MIT.
- Corpus + benchmark: CC-BY-4.0 (corpus) / CC0 (benchmark).
- Commentaries: CC-BY-4.0 (OnlineKommentar.ch) / CC-BY-SA-4.0 (OpenLegalCommentary.ch).
- Federal statutes: derived from Fedlex SPARQL endpoint; Swiss Confederation copyright applies, fair-use distribution permitted.

B English-law generalisation sketch

Goal. Demonstrate that the five-rail closing-audit decomposition is corpus-agnostic, given (a) an enumerable decision corpus, (b) a statute mirror, and (c) a citation extraction grammar for the local conventions. We sketch this for English law (UK + Welsh decisions, BAILII).

Components. Citation grammar. English neutral citations follow a fixed schema: [YEAR] COURT NN, where COURT is one of a small controlled vocabulary (UKSC, EWCA Civ, EWCA Crim, EWHC, UKHL, UKPC, EWFC, EWCOP, etc.). A regular expression captures these reliably:

```
r"\[(\d{4})\]\s+(UKSC|UKHL|UKPC|EWCA\s+Civ|EWCA\s+Crim  
|EWHC|EWCOP|EWFC|EWCC|UKEAT|UKUT|UKFTT)\s+(\d+)"
```

Statute mirror. `legislation.gov.uk` publishes machine-readable XML for every UK Public General Act since 1267. The R1 statute-existence rail can be implemented identically to our Swiss Fedlex mirror, swapping the `statutes.db` schema in place.

Decision corpus. BAILII publishes UK and Irish decisions free of charge (single-source: subscription model retired, donate-as-you-go). The corpus is approximately 750,000 decisions across UK courts. A single-script crawler is feasible.

What works without modification.

- **R1 case-citation existence:** one regex change (above).
- **R3 verbatim-quote source matching:** identical algorithm, re-uses our 5-gram shingle hashing pipeline.
- **R4 decision-date sanity:** identical ($\text{date} \leq \text{today}$ and $\geq \text{corpus floor}$).
- **R5 grounding judge:** identical (LLM-mediated, prompt templates have only the corpus name in them).

What requires adaptation.

- **R2 statute resolution:** needs the `legislation.gov.uk` URN scheme rather than Fedlex SR numbers. URN structure is well-documented; estimated 1–2 weeks of integration work.
- **Cross-lingual:** not applicable (English law is monolingual). The cross-lingual benchmark would target a different multilingual jurisdiction (Belgium, Canada-Quebec, South Africa, India).

Status. The architecture above is a sketch only; we have not yet performed an English-law evaluation. Full implementation and evaluation are left for future work. The two implementation tasks estimated above (citation-grammar swap, statute-mirror swap) are the non-trivial portions; the corpus crawl and rail orchestration reuse this paper’s harness directly.

C Cross-lingual question set

Schema. Each row is a leading case (anchored on the citation graph) with a parallel query authored in each of DE/FR/IT. The full release at `benchmarks/swiss_legal_rag_bench/cross-lingual_v1.jsonl` also carries: `q_id`, `target_lang`, `anchor_terms`, `difficulty`.

Table 10: Cross-lingual question set (page 1 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 125 V 351	accident_ insur- ance	DE	Unfallversicherungsgesetz Verbindung VwVG Beweiswürdigung Parteigutachten
BGE 125 V 351	accident_ insur- ance	FR	LAA liaison appréciation preuves expertise
BGE 125 V 351	accident_ insur- ance	IT	LAINF Cost relazione valutazione prove
BGE 134 V 231	accident_ insur- ance	DE	Unfallversicherungsgesetz Beweiswert diagnostischer Methoden funktionelle
BGE 134 V 231	accident_ insur- ance	FR	LAA valeur probante méthodes diagnostiques
BGE 134 V 231	accident_ insur- ance	IT	LAINF valore probatorio metodi diagnostici
BGE 122 V 157	constitution_ law	DE	Bundesverfassung umfassen formellen Anspruch Beizug
BGE 122 V 157	constitution_ law	FR	Cst. permettent déduire droit caractère
BGE 122 V 157	constitution_ law	IT	Cost. Cost essere dedotto diritto
BGE 130 V 343	general_ insur- ance_ law	DE	ATSG seit Januar geltenden Fassung
BGE 130 V 343	general_ insur- ance_ law	FR	LPGA teneur vigueur janvier relation
BGE 130 V 343	general_ insur- ance_ law	IT	LPGA versione vigore gennaio relazione
BGE 138 III 374	civil_ proce- dure	DE	Zivilprozessordnung Beweiserhebung Berufungsverfahren Begründung Berufung
BGE 138 III 374	civil_ proce- dure	FR	CPC administration preuves appel motivation
BGE 138 III 374	civil_ proce- dure	IT	CPC assunzione prove appello motivazione
BGE 133 II 249	federal_ court_ proce- dure	DE	Bundesgerichtsgesetz öffentlich-rechtlichen Angelegenheiten Baubewilligung Nachbarbeschwerde
BGE 133 II 249	federal_ court_ proce- dure	FR	LTF relation matière droit public
BGE 133 II 249	federal_ court_ proce- dure	IT	LTF relazione materia diritto pubblico
BGE 125	disability_ insurance	DE	Invalidenversicherungsgesetz Verbindung Mass-

Table 11: Cross-lingual question set (page 2 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 126 V 353	accident_ <u>DE</u> insur- ance	Unfallversicherungsgesetz	Arbeitswegunfall Teilzeitbeschäftigten Frage Arbeitspensum
BGE 126 V 353	accident_ <u>FR</u> insur- ance	LAA	Accident trajet victime personne
BGE 126 V 353	accident_ <u>IT</u> insur- ance	LAINF	Infortunio tragitto vittima persona
BGE 115 V 133	accident_ <u>DE</u> insur- ance	Unfallversicherungsgesetz	Präzisierung veröf- fentlichten adäquaten Kausalzusammenhang
BGE 115 V 133	accident_ <u>FR</u> insur- ance	LAA	Précision apportée publiée relative
BGE 115 V 133	accident_ <u>IT</u> insur- ance	LAINF	Precisazione concernente nesso causalità
BGE 125 V 413	disability <u>DE</u> insur- ance	Invalidenversicherungsgesetz	VwVG Verbindung Streitgegenstand Begriff
BGE 125 V 413	disability <u>FR</u> insur- ance	LAI	relation objet litige Notion
BGE 125 V 413	disability <u>IT</u> insur- ance	LAI	relazione oggetto lite Nozione
BGE 125 V 193	unemployment_ <u>DE</u> insur- ance	Arbeitslosenversicherungsgesetz	Einstellung Anspruchsberechtigung Verlust Anspruchs
BGE 125 V 193	unemployment_ <u>FR</u> insur- ance	LACI	suspension déchéance droit indemnité
BGE 125 V 193	unemployment_ <u>IT</u> insur- ance	LADI	sospensione decadenza diritto indennità
BGE 136 I 229	federal_ <u>-DE</u> court_ proce- dure	Bundesgerichtsgesetz	Anfechtung Prüfungsergeb- nisses hier Masterabschluss
BGE 136 I 229	federal_ <u>-FR</u> court_ proce- dure	LTF	contre résultat examen espèce
BGE 136 I 229	federal_ <u>-IT</u> court_ proce- dure	LTF	Cost contro esito esame
BGE 130 III 321	civil_ <u>-DE</u> code	Zivilgesetzbuch	Eintritt Versicherungsfalls Beweis Beweislast
BGE 130 III 321	civil_ <u>-FR</u> code	CC	survenance sinistre preuve Fardeau
BGE 130 III 321	civil_ <u>-IT</u> code	CC	sopravvenienza sinistro prova Onere
BGE 132 V 93	general_ <u>-DE</u> insur- ance_ law	ATSG VwVG	Begutachtung durch Sozialversicherer

Table 12: Cross-lingual question set (page 3 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 140 III 86	federal_-DE court_- proce- dure	DE	Bundesgerichtsgesetz Pflicht Begründung Rechtsver- letzungen Willenserklärung
BGE 140 III 86	federal_-FR court_- proce- dure	FR	LTF obligation motiver violations droit
BGE 140 III 86	federal_-IT court_- proce- dure	IT	LTF obbligo motivare violazioni diritto
BGE 130 V 445	general_-DE insur- ance_- law	DE	ATSG Intertemporalrechtliche Anwendbarkeit ma- terieller Bestimmungen
BGE 130 V 445	general_-FR insur- ance_- law	FR	LPGA Application intertemporelle dispositions matérielles
BGE 130 V 445	general_-IT insur- ance_- law	IT	LPGA Applicabilità intertemporale disposizioni ma- teriali
BGE 137 I 195	constitution_-DE	DE	Bundesverfassung Anspruch rechtliches Gehör Rep- likrecht
BGE 137 I 195	constitution_-FR	FR	Cst. droit être entendu réplique
BGE 137 I 195	constitution_-IT	IT	Cost. Cost diritto essere sentito
BGE 140 III 115	international_-DE pri- vate_- law	DE	IPR-Gesetz zweiter Spiegelstrich LugÜ Interna- tionale
BGE 140 III 115	international_-FR pri- vate_- law	FR	LDIP second tiret Compétence internationale
BGE 140 III 115	international_-IT pri- vate_- law	IT	LDIP secondo trattino CLug Competenza
BGE 129 I 8	constitution_-DE	DE	Bundesverfassung Genfer Reglements April über
BGE 129 I 8	constitution_-FR	FR	Cst. Règlement genevois avril fixant
BGE 129 I 8	constitution_-IT	IT	Cost. Cost Regolamento ginevrino aprile
BGE 143 IV 241	criminal_-DE proce- dure	DE	Strafprozessordnung StPO häusliche Gewalt Einstel- lung
BGE 143 IV 241	criminal_-FR proce- dure	FR	CPP violence domestique classement pénale
BGE 143 IV 241	criminal_-IT proce- dure	IT	CPP violenza domestica abbandono procedimento
BGE 141	criminal_-DE	DE	Strafgesetzbuch StGB Mord Konkurrenz mehreren

Table 13: Cross-lingual question set (page 4 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 144 V 210	misc	DE	misc Verordnung Europäischen Parlaments Rates
BGE 144 V 210	misc	FR	misc Règlement Parlement européen Conseil
BGE 144 V 210	misc	IT	misc Regolamento Parlamento europeo Consiglio
BGE 136 IV 55	criminal_DE code		Strafgesetzbuch StGB Strafzumessung vermindelter Schuldfähigkeit
BGE 136 IV 55	criminal_FR code		CP fixation peine responsabilité restreinte
BGE 136 IV 55	criminal_IT code		CP commisurazione pena caso scemata
BGE 142 III 413	civil_- proce- dure	DE	Zivilprozessordnung Novenrecht Berufungsinstanz Grundsätze Berufungsverfahrens
BGE 142 III 413	civil_- proce- dure	FR	CPC recevabilité nova appel Principes
BGE 142 III 413	civil_- proce- dure	IT	CPC nova sede appello Principi
BGE 110 V 48	misc	DE	misc Anfechtungsgegenstand Streitgegenstand Ver- fügung Sachurteilsvoraussetzung
BGE 110 V 48	misc	FR	misc Objet contestation litige existence
BGE 110 V 48	misc	IT	misc Oggetto impugnato litigioso esistenza
BGE 137 III 617	civil_- proce- dure	DE	Zivilprozessordnung Berufungsanträge Berufung- seingabe Anträge enthalten
BGE 137 III 617	civil_- proce- dure	FR	CPC conclusions appel mémoire doit
BGE 137 III 617	civil_- proce- dure	IT	CPC conclusioni appello allegato deve
BGE 129 V 1	old_- age_- insur- ance	DE	AHVG Ende gültig gewesenenen Fassung
BGE 129 V 1	old_- age_- insur- ance	FR	LAVS teneur vigueur jusqu corrélation
BGE 129 V 1	old_- age_- insur- ance	IT	LAVS tenore vigente fino dicembre
BGE 126 I 97	criminal_DE code		Strafgesetzbuch BStP Zulässigkeit Geschädigten gegen
BGE 126 I 97	criminal_FR code		CP Recevabilité droit public interjeté
BGE 126 I 97	criminal_IT code		CP Ammissibilità diritto pubblico inoltrato
BGE 143 III 65	civil_- code	DE	Zivilgesetzbuch Anhörung Pflegeeltern Entlassung Beiständin
BGE 143 III 65	civil_- code	FR	CC audition parents nourriciers libération

Table 14: Cross-lingual question set (page 5 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 144 IV 345	criminal_ proce- dure	DE	Strafprozessordnung StPO Unschuldsv Vermutung Be- weiswürdigung Wiederholung
BGE 144 IV 345	criminal_ proce- dure	FR	CPP présomption innocence appréciation preuves
BGE 144 IV 345	criminal_ proce- dure	IT	CPP Cost presunzione innocenza valutazione
BGE 128 III 411	civil_ code	DE	Zivilgesetzbuch Kinderunterhaltsbeitrag Unter- suchungsmaxime Tragweite Diese
BGE 128 III 411	civil_ code	FR	CC contribution entretien faveur enfants
BGE 128 III 411	civil_ code	IT	CC contributo mantenimento prole massima
BGE 146 IV 88	criminal_ proce- dure	DE	Strafprozessordnung Verweigerung Massnahmen Feststellung Fahruntfähigkeit
BGE 146 IV 88	criminal_ proce- dure	FR	CPP lien refus mesures constatation
BGE 146 IV 88	criminal_ proce- dure	IT	CPP unitamente LCStr rifiuto sottoporsi
BGE 144 III 349	civil_ proce- dure	DE	Zivilprozessordnung Zulässigkeit Noven Beru- fungsverfahren uneingeschränkte
BGE 144 III 349	civil_ proce- dure	FR	CPC recevabilité nova appel maxime
BGE 144 III 349	civil_ proce- dure	IT	CPC ammissibilità nova appello principio
BGE 129 I 129	old_ court_ law	DE	OG Anspruch unentgeltlichen Rechtsbeistand an- fechtbarer
BGE 129 I 129	old_ court_ law	FR	OJ droit être assisté gratuitement
BGE 129 I 129	old_ court_ law	IT	OG Cost diritto avvocato ufficio
BGE 142 I 135	foreigners_ law	DE	AuG Zulässigkeit öffentlich-rechtlichen Angelegen- heiten gegen
BGE 142 I 135	foreigners_ law	FR	AuG LEtr admissibilité matière droit
BGE 142 I 135	foreigners_ law	IT	AuG Cost LStr ammissibilità materia
BGE 138 I 232	contract_ tort	DE	Obligationenrecht Personalreglement Öffentlichen Verkehrsbetriebe Genf
BGE 138 I 232	contract_ tort	FR	CO Statut personnel Transports publics
BGE 138 I 232	contract_ tort	IT	CO Cost segg Statuto personale
BGE 130 II 281	echr	DE	EMRK Familiennachzug gefestigtes Anwesenheits- recht kombinierten
BGE 130 II 281	echr	FR	CEDH regroupement familial droit présence